Software Engineering & Project Management (SEPM)

Energy & Utility Management System (EUMS)

Billing, metering and customer service for electricity, water and gas

Use Cases and User Stories

Software process model: Waterfall · Combined SEPM + DBMS course project

Group number	Group 3
Course and section	Software Engineering & Project Management (SEPM), CSEF
Date	October 2026

#	Name	Registration number
1	Sumit P Surendran	20243590
2	Sreethan Reddy	20243550
3	Sushrut Lad	20243551
4	Viswajith	20243559

1. Requirements Model: Use Cases and User Stories

1.1 Use-case diagram

Three actors use the system. "Employee" covers the meter reader, billing officer and technician, who share one role in the system. UC-05 *Generate bill* is never started by a person; it is included in UC-04, so every saved reading produces a bill.

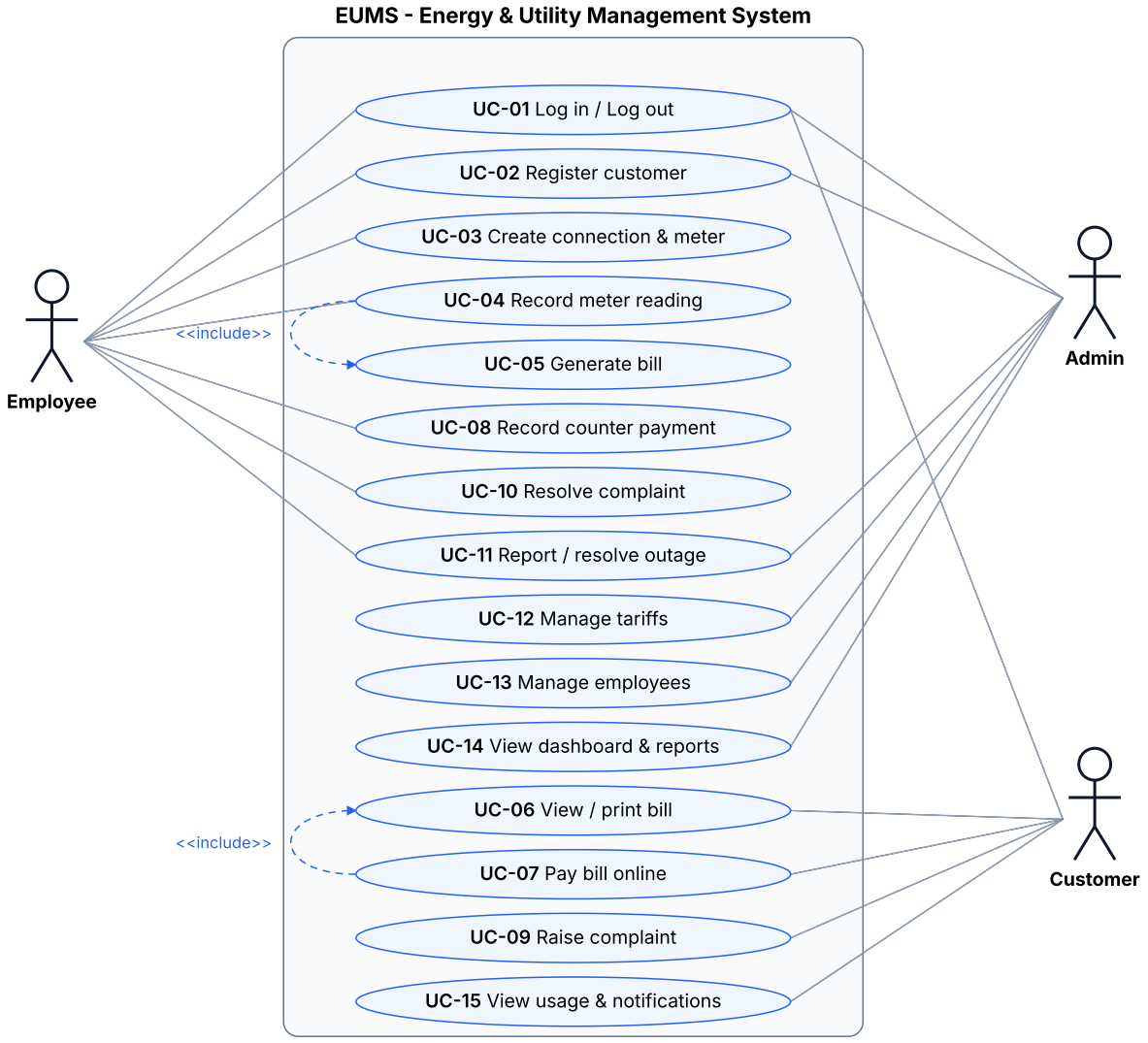


Figure 1: Use-case diagram of EUMS

1.2 Use-case descriptions

UC-04 Record meter reading (includes UC-05 Generate bill)	
Actor(s)	Employee
Pre-condition	Employee is logged in; the connection and meter are active.
Main flow	1. Employee opens Meter Readings → Add Reading. 2. Employee selects a meter; system shows customer name, utility and previous reading with its date. 3. Employee types the current reading; system shows the consumption immediately. 4. Employee enters the reading date and optional remarks and clicks Save Reading. 5. System saves the reading, calculates the bill from the tariff slabs (UC-05), saves the bill and its items, and notifies the customer. 6. System opens the new bill.
Alternate / exception flows	• 3a. Current reading lower than previous: an error is shown and Save is disabled. If submitted anyway, the database trigger rejects it and nothing is saved. • 4a. Date on or before the last reading date: the trigger rejects it and an error is shown. • 4b. Reading already exists for that date: error shown.
Post-condition	Reading and bill stored; customer sees the bill and a "New bill generated" notification.

UC-07 Pay bill online	
Actor(s)	Customer
Pre-condition	Customer is logged in and has a bill that is unpaid or partial.
Main flow	1. Customer opens Bills & Payments and clicks Pay on a bill. 2. System shows bill details and the outstanding amount, pre-filled. 3. Customer optionally changes the amount (partial payment) and chooses card / UPI / net banking / wallet. 4. Customer clicks Pay. 5. System records the payment with a receipt number; trigger updates bill to paid or partial and notifies the customer. 6. System shows the bill with the payment and a success message.
Alternate / exception flows	• 1a. Bill belongs to another customer: HTTP 403. • 4a. Amount more than outstanding: trigger rejects; error shown; nothing saved. • 4b. No method chosen: error shown. • 1b. Bill already paid: message "already paid".
Post-condition	Payment saved; bill status updated; receipt number shown.

UC-09 / UC-10 Raise and resolve complaint	
Actor(s)	Customer, Employee
Pre-condition	Users are logged in.
Main flow	1. Customer opens Complaints, chooses connection, types the issue, details and urgency, and submits. 2. System creates complaint CMPxxxx with status Open. 3. Employee opens Complaints, filters Open, and opens the complaint. 4. Employee assigns a technician, sets priority, adds notes and changes status to In Progress or Resolved. 5. System saves the change, records resolution time, and notifies the customer through a trigger.
Alternate / exception flows	• 1a. Issue left empty: error shown.
Post-condition	Complaint tracked to resolution; customer informed of each status change.

UC-11 Report outage	
Actor(s)	Admin, Employee
Pre-condition	User is logged in as staff.
Main flow	1. Staff opens Outages → Report Outage. 2. Staff selects utility and area and enters reason, start time and estimated end. 3. System saves the outage as Active (or Scheduled if it starts in the future). 4. Trigger notifies every customer in that area who has that utility. 5. Later, staff clicks Mark resolved.
Alternate / exception flows	• 2a. End time not after start time: error shown.
Post-condition	Outage visible to staff and affected customers.

UC-01 Log in	
Actor(s)	All
Pre-condition	User has an active account.
Main flow	1. User opens the site and enters username/email and password. 2. System checks the password hash. 3. System opens the dashboard for the user's role.
Alternate / exception flows	• 2a. Wrong credentials: "Invalid username or password". • Any page opened without login redirects to the login page.
Post-condition	User session created.

1.3 User stories with acceptance criteria

ID	User story	Acceptance criteria	Status
US-01	As a **meter reader**, I want to see the previous reading when I pick a meter, so that I do not mistype the new one.	Given a meter is selected, previous value and date are shown; consumption updates while typing; lower values show an error and disable Save.	Done
US-02	As a **billing officer**, I want bills to be calculated automatically from slabs, so that there are no manual errors.	For 250 kWh on the domestic plan the bill is ₹1,207.50 with two slab rows, fixed charge and 5% tax.	Done
US-03	As a **customer**, I want to pay my bill online by UPI or card, so that I do not have to visit the office.	Pay page shows outstanding amount; payment creates receipt RCPyyyymmdd-nnnn; bill becomes Paid; overpayment rejected.	Done
US-04	As a **customer**, I want to see all my connections and current bills on one dashboard.	Dashboard shows one card per connection with latest bill amount, due date and status, plus a 6-month chart.	Done
US-05	As a **customer**, I want to be told about outages in my area.	When staff report an outage for area A and utility U, all customers in A with U get a notification; others do not.	Done
US-06	As a **technician**, I want to see open complaints and update them.	Complaints page has tabs All/Open/In progress/Resolved with counts; saving Resolved records the time and notifies the customer.	Done
US-07	As an **admin**, I want to change tariff rates without changing old bills.	Edited rates are used for the next bill; previously issued bill totals are unchanged.	Done
US-08	As an **admin**, I want to see revenue, dues and complaints at a glance.	Dashboard shows 6 KPI cards, billed amount by utility for 6 months, collections split and recent activity.	Done
US-09	As an **admin**, I want overdue reminders and late fees to be applied automatically.	Bills unpaid after the due date get a surcharge and the customer is reminded.	Planned
US-10	As a **billing officer**, I want to replace a faulty meter without breaking reading history.	Old meter marked Replaced; new meter starts at its own baseline; next bill uses the new meter.	Planned

Table 1: User stories